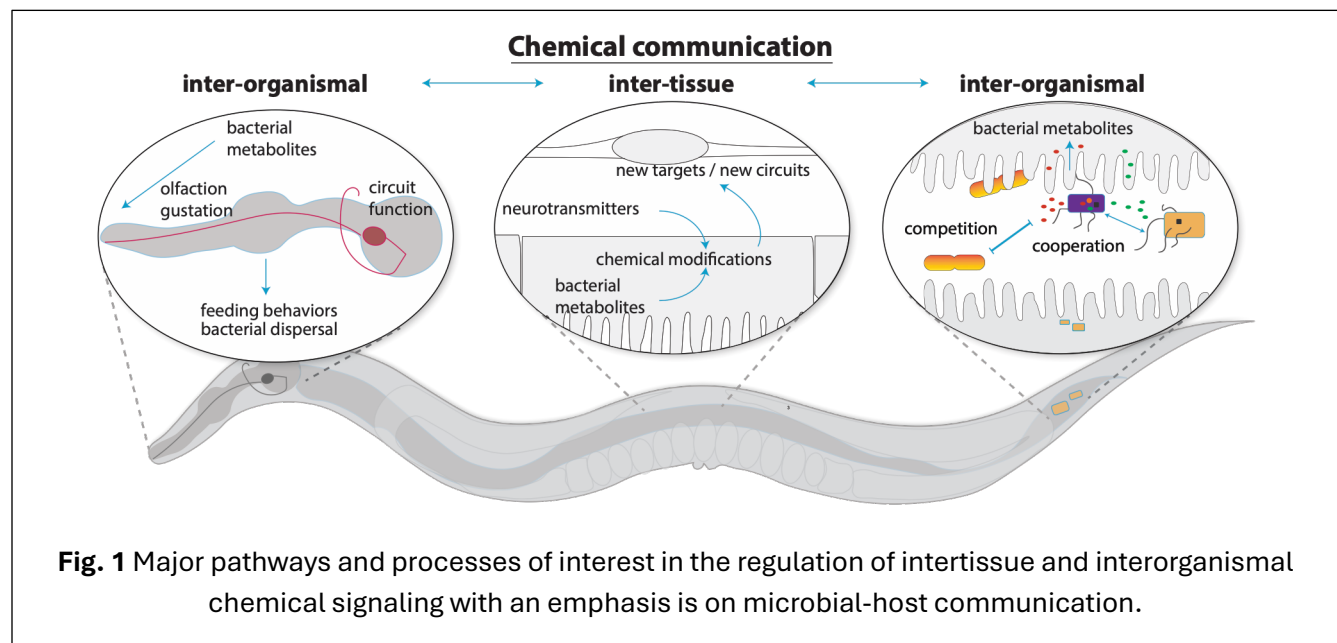


Laboratory of intertissue and interorganismal chemical communication

Animal nervous systems evolved in environments surrounded by microbes. The earliest animals were, and many extant species are, microbe feeders. Our lab is motivated by the thesis that A) chemicals from microorganisms are critical for proper animal behavior and physiology and B) these interorganismal signals played a major role in the origins of neuronal communication. This is an understudied area, because identifying these signals within the complex chemical milieu found in most host-microbe communities remains challenging. Several key questions are largely unresolved in most host-microbe systems, such as (1) what are the relevant microbial chemical signals and how are these signals modified or trafficked within the host, (2) how (or if) hosts perceive and respond to intestinal bacteria, and (3) how (or if) host feeding decisions shape gut microbial communities. In my lab we leverage the many tools available in *C. elegans* to identify novel mechanisms of host-microbe and inter-tissue communication. We combine bacterial and worm genetics, high-throughput screening, behavioral analysis, metabolomics and high-resolution neural activity imaging. Below, I summarize my lab's key advances and highlight my perspective on the future of this field.

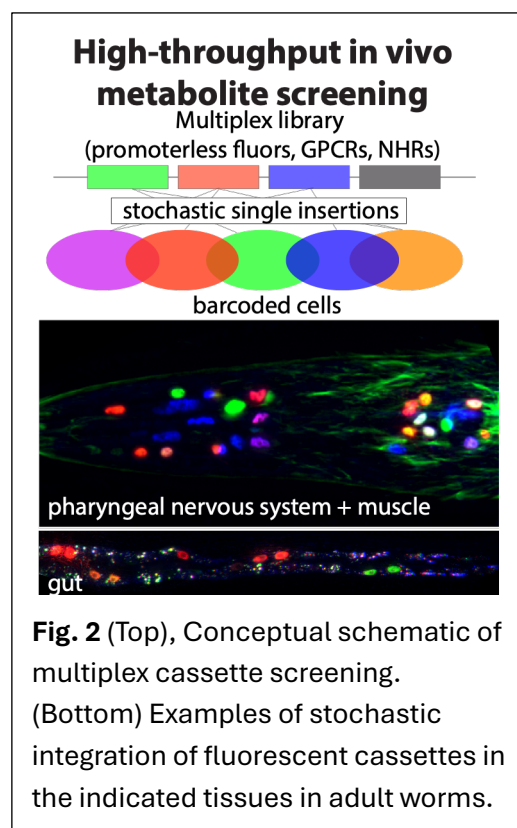
**Direction 1: Intertissue trafficking and modification of gut metabolites.**

I previously showed that gut microbial production of neurotransmitters alters chemosensory behaviors in *C. elegans* (O'Donnell et al., *Nature* 2020). In an ongoing collaboration with the Schroeder lab (Cornell), we discovered that both microbe-produced and host-produced neurotransmitters are glucosylated in the gut – a potential mechanism for transmitter reuse

([Wrobel et al., JACS 2021](#)). Our recent work, in revision at *Nature* ([Malaiwong et al., 2026](#)), demonstrates that octopamine neurotransmitter glucosylation in the intestine leads to modification with specific fatty acids that encode information about worm dietary history and metabolism. We identified an intertissue pathway and several conserved biosynthetic and transport factors, leading to hydrolysis of glucosylated octopamine and chemosensory behavior changes. We've subsequently identified additional glucosylated microbe- and host-derived metabolites that are trafficked between tissues to alter behaviors. We plan to ascertain the effects of these metabolites on neural circuit function using our toolkit of genetic and neuronal imaging techniques (Fig. 1 and [Campagna et al., bioRxiv 2026](#), [Philbrook et al., PLOS Bio 2024](#), [Khan et al., PLOS Bio 2022](#)).

Direction 2: Mechanisms of interorganismal chemical signal production and sensation

Natural microbial communities are complex and depend on specific niches, which can limit the use of traditional high-throughput cultured cell-based receptor-ligand screening methods (e.g. Presto-Tango/Salsa)^{1,2}. In my lab, we've developed new methods for rapid production of complex, multicopy integrated transgenic arrays in *C. elegans* ([Malaiwong et al., 2026 in revision at G3](#)). We've used these techniques to overcome a significant barrier to receptor-ligand screening in living multicellular animals: the ability to simultaneously express different receptors in individual cells in adult animals and use barcodes to mark cells activated by ligands (Fig. 2). We will identify microbe-produced GPCR agonists, with emphasis on monoamine and opioid receptor systems, which regulate satiety and feeding. In a collaboration with the Musser lab (Yale), we found that monoamine signaling may have an ancient origin that predates nervous systems (Zang et al., *bioRxiv* 2026). Based on our work and others' demonstrating that gut bacteria³ and other single-celled organisms⁴ can produce these transmitters, we theorize that these were key interorganismal signals before their adoption in nervous system signaling.



Direction 3: Coordination of feeding behaviors and predator-prey dynamics

We showed that intestinal colonization by tyramine-producing bacteria subsequently increases worms' association with, and feeding on, these bacteria, suggesting that there may be tradeoffs between benefits of gut colonization (e.g. nutrient-rich environment and dispersal advantage) and

predation³. Microbes can survive in the *C. elegans* intestine and be dispersed via surface contact or defecation^{3,5,6}. We are studying how feeding is altered by chemical signals from these microbes, and how host foraging impacts bacterial dispersal and community dynamics. In collaboration with the Brown lab (ICL), we screened a subset of our nematode-associated bacteria (~100 strains) using high-throughput analysis of worm locomotion and identified behavioral signatures spanning various taxa. Using this collection, we are studying the dynamics of predation, associative learning and bacterial dispersal using *C. elegans* and, in collaboration with the Yan lab (Yale), using microbe-feeding *Tetrahymena* protists. The main work in directions 2 and 3 are currently supported by a NIH Director's New Innovator award.

Future plans and perspectives

My lab's thesis is that animal and microbe systems are dynamically organized via extensive chemical communication. We continue to develop a comprehensive picture of the key chemical and trafficking steps for host detection of microbial cues (both externally and internally), and how chemical signaling-dependent behavior shapes microbial community dynamics. In the near term, our work is focused on *C. elegans* due to the extensive tools available in this system, but we expect that many of the "rules" we discover using this system will apply to other animals, which we plan to explore in the long term.

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